

Test 2 Review

NAME: _____

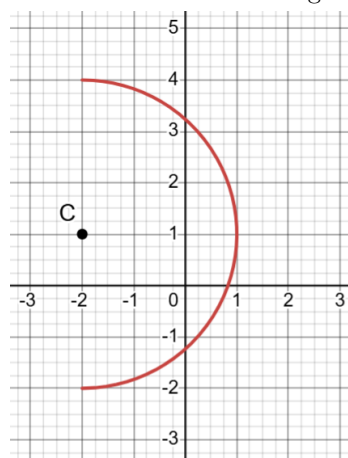
Q.1. An airplane flying north at 200 mi/hr passed over a point on the ground at 2 : 00 P.M. Another airplane at the same altitude passed over the point at 2 : 30 P.M., flying east at 400 mi/hr. If t denotes the time in hours after 2 : 30 P.M., express the distance d between the airplanes in terms of t .

Q.2. Two square wire frames are to be constructed from a piece of wire 100 inches long. If the area enclosed by one frame is to be one-half the area enclosed by the other, find the dimensions of each frame. (Disregard the thickness of the wire.)

Q.3. Show that the triangle with vertices $A(3, 1)$, $B(-5, -3)$, and $C(4, -1)$ is a right triangle, and find its area.

Q.4. Given $A(-1, 2)$ and $B(3, -4)$, find a general form of an equation for the perpendicular bisector of segment AB .

Q.5. Find the equation of the semicircle whose center is given below:



Q.6. Find the center and radius of the circle with the following equation in the $x - y$ plane:
 $x^2 + y^2 + 4x - 16y + 64 = 0$.

Q.7. Compute the difference quotient, $\frac{f(a+h)-f(a)}{h}$, for the function $f(x) = \frac{1}{x+2}$ (you may take $a, a+h \neq -2$ and $h \neq 0$).

Q.8. Find an equation of the line through $A\left(\frac{1}{2}, -\frac{1}{3}\right)$ that is (a) parallel to the line $6x+2y+5=0$ and (b) perpendicular to the line $6x+2y+5=0$.

Q.9. Is the following function even, odd, or neither? $f(x) = \sqrt[3]{x^4 + 3x^2 + 5}$

Q.10. Sketch the graph of the following function and state the intervals on which it is increasing, decreasing, and constant, respectively:

$$f(x) = \begin{cases} x^2 : & \text{if } x < 0 \\ 3x : & \text{if } 0 \leq x < 2 \\ 6 : & \text{if } x > 2 \end{cases}$$

Q.11. Find the minimum or maximum of the following function $f(x) = -3x^2 + 30x - 82$.

Q.12.

78 Distance between cars At noon, car A is 10 feet to the right and 20 feet ahead of car B, as shown in the figure. If car A continues at 88 ft/sec (or 60 mi/hr) while car B continues at 66 ft/sec (or 45 mi/hr), express the distance d between the cars as a function of t , where t denotes the number of seconds after noon.

Exercise 78

